A Bayesian rating with winning margins

The CueVR World Ranking and game rankings

CueVR

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Abstract

We rate players of several cue-sport games from verified online frames. Each player has one latent skill. Each family of games has its own scale on which skill shows in who wins, and its own linear-Gaussian model for the winning margin. Skills and family parameters are estimated jointly, from the whole history at once, as a maximum a posteriori fit by block coordinate ascent. A frame's margin is allowed to tell no more about the skill gap than its result does. Uncertainties come from the inverse of the full Hessian, so a long record against a few opponents is not taken for a well-placed one. Each frame was predicted from the frames before it, with the model refitted before every frame. On 1495 frames scored this way, the model reached a mean log loss of 0.6625, against 0.6713 for the incremental Glicko-style rating used before it (3.1 standard errors). Using the margin, rather than the outcome alone, improves the log loss by 3.2 standard errors.

Status. Since 1 November 2026 this model is the CueVR World Ranking and each game's ranking in the club, with each player's rating shown beside their name (the conservative score of Section 4, on the scale described there). It replaced ClaudeELO for the World Ranking and CUELO for the game rankings. The month's leader at the end of each month wins that ranking's trophy.

1 Data

Frames $n = 1, \dots, N$ are verified singles frames between two proved accounts, with a winner. Frames whose two ends share an internet address are excluded. Each frame has a family $f(n) \in \mathcal{F} = \{\text{8-ball, 9/10-ball, snooker, billiards, straight pool, other}\}$, a winner $w(n)$ and a loser $l(n)$. Let $\mathcal{M} \subseteq \{1, \dots, N\}$ be the frames that also have a *margin* $d_n \in [-1, 1]$, measured from the winner's side and clipped to $[-1, 1]$:

Family	d_n
8-ball	$(\text{pots}_w - \text{pots}_l)/7$
9/10-ball	$(\text{pots}_w - \text{pots}_l)/9$
snooker, R reds	$(\text{score}_w - \text{score}_l)/(8R + 27)$
billiards, straight pool	$(\text{score}_w - \text{score}_l)/\max(\text{score}_w, \text{score}_l)$

A forfeited frame counts as a win for the player who stayed, but has no margin. Games in "other" have no margin either.

2 Model

Each player i has a skill $\theta_i \in \mathbb{R}$. Each family f has an outcome scale β_f, a margin slope κ_f and a margin variance σ_f^2. Write $x_n = \theta_{w(n)} - \theta_{l(n)}$ and $\Lambda(u) = 1/(1+e^{-u})$. Then

$$\Pr(w(n) \text{ beats } l(n) \mid \theta, \beta) = \Lambda(\beta_{f(n)}\, x_n), \tag{1}$$

$$d_n \mid \theta, \kappa, \sigma \sim \mathcal{N}\big(\kappa_{f(n)}\, x_n,\ \sigma^2_{f(n)}\big), \qquad n \in \mathcal{M}, \tag{2}$$

with the outcome and the margin conditionally independent given θ. The margin term enters the fit tempered, with a weight $\omega_n \in (0, 1]$ (Section 2.1). The priors are

$$\theta_i \sim \mathcal{N}(0, \tau^2), \quad \tau = 0.5; \qquad \beta_f \sim \mathcal{N}(1, 0.3^2); \qquad \kappa_f \sim \mathcal{N}(0, 1). \tag{3}$$

σ_f^2 is estimated by shrinkage toward $s_0^2 = 0.3^2$ with the weight of $\nu = 2$ frames (Section 3). With every $\kappa_f = 0$ the model reduces to a Bradley–Terry model with a separate discrimination for each family. The margin term is a Gaussian margin-of-victory component, in the spirit of Harville and Glickman.

2.1 The margin cap

Per frame, the outcome carries Fisher information $I_n^{\mathrm{o}} = \beta_f^2 p_n(1-p_n) \le \beta_f^2/4$ about x_n, and the margin carries $I_f^{\mathrm{m}} = \kappa_f^2/\sigma_f^2$, the same for every frame of the family. Left alone, the margin outweighed the result by up to 2.5 times, whatever the odds; a long run of heavy wins over weaker players then moved a rating as if each were an upset. Each margin's log-likelihood term is therefore multiplied by

$$\omega_n = \min\Big(1,\ \frac{\beta_f^2\, p_n(1-p_n)}{\kappa_f^2/\sigma_f^2}\Big), \tag{4}$$

so that the margin tells at most what the result tells, and least where the result was expected (p_n near 0 or 1). This is a tempered (power) likelihood, not a change to the model of d_n; ω_n is evaluated at the current fit and held fixed within each Newton step. On the present data $I_f^{\mathrm{m}} > \beta_f^2/4$ in every family, so the cap binds on every frame with a margin and each margin counts exactly as much as its result. The mean weight is 0.56 in 8-ball, 0.35 in billiards, 0.25 in snooker, 0.16 in 9/10-ball and 0.08 in straight pool.

3 Estimation

We take the MAP estimate of (θ, β, κ). For fixed σ^2, the unnormalised log posterior is

$$\mathcal{L} = \sum_n \log \Lambda(\beta_f x_n) - \sum_{n\in\mathcal{M}} \omega_n \frac{(d_n - \kappa_f x_n)^2}{2\sigma_f^2} - \sum_i \frac{\theta_i^2}{2\tau^2} - \sum_f \frac{(\beta_f - 1)^2}{2\cdot 0.3^2} - \sum_f \frac{\kappa_f^2}{2}, \tag{5}$$

where f stands for $f(n)$ inside each sum over frames. Write $p_n = \Lambda(\beta_f x_n)$, and let $s_{ni} = +1$ if player i won frame n and -1 if they lost it. Each sweep of the block coordinate ascent updates, in turn:

1. **Skills.** For each player i, one Newton step $\theta_i \leftarrow \theta_i + g_i/h_i$, with

$$g_i = -\frac{\theta_i}{\tau^2} + \sum_{n\ni i} s_{ni}\, \beta_f (1-p_n) + \sum_{n\ni i,\ n\in\mathcal{M}} s_{ni}\, \omega_n \frac{\kappa_f (d_n - \kappa_f x_n)}{\sigma_f^2},$$

$$h_i = \frac{1}{\tau^2} + \sum_{n\ni i} \beta_f^2\, p_n(1-p_n) + \sum_{n\ni i,\ n\in\mathcal{M}} \omega_n \frac{\kappa_f^2}{\sigma_f^2}.$$

The skill is updated in place, so later players in the sweep see it (Gauss–Seidel).

2. **Outcome scales.** For each family f, one Newton step on β_f, with

$$g = -\frac{\beta_f - 1}{0.3^2} + \sum_{n \in f} x_n(1 - p_n), \qquad h = \frac{1}{0.3^2} + \sum_{n \in f} x_n^2\, p_n(1 - p_n).$$

3. **Margin slopes**, the exact conditional maximiser. The family parameters use every margin unweighted; the cap applies only to what a margin says about the players:

$$\kappa_f = \frac{\sum_{n \in f \cap \mathcal{M}} x_n d_n / \sigma_f^2}{\sum_{n \in f \cap \mathcal{M}} x_n^2 / \sigma_f^2 + 1}.$$

4. **Margin variances**, shrunk toward s_0^2:

$$\sigma_f^2 = \frac{\sum_{n \in f \cap \mathcal{M}} (d_n - \kappa_f x_n)^2 + \nu s_0^2}{|f \cap \mathcal{M}| + \nu}.$$

The sweeps stop when the largest update is below 10^{-5}, or after 300 sweeps. With this tolerance every reported rating is within 0.04 points of the exact optimum. When new results arrive, the fit is warm-started from the previous solution. The server works the rankings out again at most once every five minutes; results arriving inside that wait are marked pending and taken in when it ends. On a desktop, at 1723 frames, the full listing with its uncertainties takes about 190 ms and the game rankings about 80 ms more (CUELO 57 ms, Elo 2 ms).

4 What is reported

The rating and its uncertainty are

$$R_i = 1500 + \frac{400}{\ln 10}\,\hat{\theta}_i, \qquad \mathrm{SD}_i = \frac{400/\ln 10}{\sqrt{H_{ii}}}, \tag{6}$$

where $(H^{-1})_{ii}$ is the ith diagonal entry of the inverse of the negative Hessian H of the skills at the mode (a Laplace approximation). With $c_n = \hat{\beta}_f^2 \hat{p}_n(1 - \hat{p}_n) + \omega_n \hat{\kappa}_f^2 / \hat{\sigma}_f^2$ (ω_n only for $n \in \mathcal{M}$),

$$H = \frac{1}{\tau^2} I + \sum_n c_n \,(e_{w(n)} - e_{l(n)})(e_{w(n)} - e_{l(n)})^\top, \tag{7}$$

a prior term plus a weighted graph Laplacian of who has played whom. The first version reported $1/\sqrt{H_{ii}}$, each player's own information alone. That understates the uncertainty whenever a player's frames are against few opponents, because many frames settle the gap *to those opponents*, not the place in the field: one player with 22 US 8-ball frames against three others was shown ±24 points, and is ±40 by the full inverse.

The diagonal of H^{-1} is computed by a Cholesky factorisation (numpy/LAPACK; 71 ms at 1000 players). Without numpy, a plain-Python dense inverse is used up to 300 players; beyond that each needed entry $(H^{-1})_{ii}$ is found by conjugate gradients on the sparse H, at the cost of a few products with H, each linear in the number of pairings. Since $1/H_{ii} \le (H^{-1})_{ii}$, the cheap figure can only flatter a player; the exact one is computed best first, by the flattering score, until no remaining player could reach the places shown. The sparse and dense results agree.

Players are ranked by the conservative score $R_i - 2\,\mathrm{SD}_i$. A player is *provisional* with fewer than 20 frames or fewer than 5 different opponents. On this scale, 400 points means odds of 10 to 1 in a family with $\beta_f = 1$. A separate tool draws from the full Laplace covariance to give each player's chance of being ranked first, or in the top three.

The rankings. The World Ranking lists, by conservative score, every player with at least 15 frames and 4 different opponents. Each game's ranking is the same model fitted on that game group's frames alone, listing players with at least 10 frames and 4 different opponents.

5 Fitted family parameters

Table 1 shows the fit, with the cap, on all 1723 rated frames (1638 with a margin) from 10 to 26 October 2026. Per frame, the information about x_n from the outcome is $\beta_f^2 p_n(1-p_n) \le \beta_f^2/4$, and from the margin it is κ_f^2/σ_f^2. In every family with a margin, the margin on its own would carry more than the result; in 9/10-ball, where the outcome is close to a coin toss, about six times as much. The cap brings each margin down to its result's share. Against the uncapped fit of 24 October, κ_f is a little lower in every family (snooker 0.45 to 0.38).

Family	Frames	$\hat{\beta}_f$	$\hat{\kappa}_f$	$\hat{\sigma}_f$	outcome info ($\le$)	margin info
8-ball	679	1.73	0.52	0.48	0.75	1.16
snooker	230	1.27	0.38	0.30	0.40	1.55
billiards	92	1.24	0.45	0.45	0.39	1.02
straight pool	28	1.20	0.60	0.30	0.36	4.02
9/10-ball	655	0.68	0.32	0.39	0.11	0.68
other	39	0.93	—	—	0.22	—

Table 1: Family parameters fitted on all frames, with the cap. "Margin info" is before the cap. Straight pool rests on 28 frames.

6 Evaluation

Each frame from 13 October onward (1495 of the 1723 frames) was predicted using only the frames before it, with the model refitted before every frame. Table 2 compares the models' mean log loss; a coin toss scores $\ln 2 \approx 0.6931$. Standard errors are of the mean per-frame difference. "Picks" is the share of frames whose winner was given better than even odds.

Model	Log loss	Picks	vs. CUELO
Elo ($K = 30$, all games)	0.6812	56.9%	−2.9 SE
CUELO (Glicko-style, per-game weights)	0.6713	58.7%	—
ClaudeELO (CUELO plus a play-figures bonus)	0.6716	59.2%	−0.2 SE
ClaudeELO 2.0	0.6649	57.7%	+2.4 SE
This model, outcome only, refitted daily	0.6664	58.3%	+1.6 SE
This model, outcome only ($\kappa \equiv 0$)	0.6658	58.1%	+2.2 SE
This model, margins uncapped	0.6611	57.8%	—
This model, margins capped (as run)	**0.6625**	57.9%	**+3.1 SE**

Table 2: One-step-ahead log loss over 1495 frames. The capped margin improves on the outcome-only model by 3.2 SE per frame. The uncapped margin scores 1.2 SE better than the capped one.

The cap costs a little predictive accuracy (0.0014 in log loss, 1.2 SE) and is kept on purpose. Predicting the next frame rewards trusting the margin fully; ranking players should not let a run of expected heavy wins count as more than the wins. By family, the capped model's log loss is 0.6342 in 8-ball, 0.6909 in 9/10-ball, 0.6648 in snooker and 0.6389 in billiards: the best of all the models in 8-ball, snooker and billiards, and 0.005 behind ClaudeELO 2.0 in 9/10-ball.

7 Limitations

Skill is static: a frame from the first week counts as much as one from yesterday, and a player who stops playing keeps their place and their certainty. The two margin definitions for pool count balls, not difficulty. The cap is a choice about what a ranking should reward, not a fitted quantity. Seventeen days of play is little to judge any of this on.

8 Future work: skill that changes over time

Model. Replace θ_i by $\theta_{i,t}$, one value per player per day t on which they played, linked by a Gaussian random walk:

$$\theta_{i,t_1} \sim \mathcal{N}(0, \tau^2), \qquad \theta_{i,t_{k+1}} \mid \theta_{i,t_k} \sim \mathcal{N}\big(\theta_{i,t_k},\ \delta^2\,(t_{k+1} - t_k)\big), \tag{8}$$

with frame n on day t using $x_n = \theta_{w(n),t} - \theta_{l(n),t}$ in both the outcome and the (capped) margin terms. This is Whole-History Rating (Coulom, 2008) and TrueSkill Through Time (Dangauthier et al., 2007). With $\delta = 0$ it is the present model. The family parameters stay static.

Fitting. The MAP fit keeps the same block coordinate ascent. For one player the Hessian over their own days is tridiagonal: the frame terms on the diagonal, and $\pm 1/(\delta^2 \Delta t)$ from the random walk linking neighbouring days. A Newton step for the whole of a player's history is a tridiagonal solve, linear in the number of days they played, so a sweep still costs about the number of frames. Warm starts work as now.

Uncertainty and inactivity. A player's reported skill is their value on their last day, carried forward to today: its variance is the posterior variance on that day plus δ^2 times the days since. The SD grows while they are away, so their conservative score $R_i - 2\,\mathrm{SD}_i$ drifts down until they play again, and a returning player's first frames move them quickly. The whole-fit uncertainty of Section 4 then needs the block-tridiagonal structure in time inside the sparse Hessian; conjugate gradients carry over directly, and a selected inverse (block tridiagonal) gives each player's last-day variance.

Choosing δ. By the same one-step-ahead log loss as Table 2, over a grid of δ including 0, keeping $\delta = 0$ unless a larger value wins by a clear margin. With weeks of data the answer may well be 0; the drift becomes measurable over months.

A cheaper first step. Weight each frame's log-likelihood by $2^{-a_n/h}$ for a frame a_n days old, with the half-life h chosen by the same evaluation. This needs no new fitting code, but it is not a model of change: it does not let uncertainty grow with absence, and it discards information rather than reinterpreting it.

The method in plain terms

One number per player. Every player has a single hidden skill level, and the system estimates it from their verified online frames. Everyone starts at the same level and moves away from it only as their results justify. A handful of early lucky wins is not enough to make a big difference.

How you won matters, within limits. The system looks at the score as well as the result. Winning a frame comfortably says more about the gap between two players than winning it narrowly. But a big win counts for no more than the win itself, and least of all when the result was expected. A strong player beating a much weaker one by a lot is what everyone would predict, so it changes little.

Some games are more predictable than others. In snooker the better player usually wins, so each snooker frame says a lot about skill. In 9-ball luck plays a bigger part, so each frame says less. The system is not told this; it measures it from the results.

The whole history is reconsidered. The ratings are worked out from every frame at once, not one result at a time. If you lose to a newcomer who later proves to be one of the best players, that loss is reassessed as no great surprise and your rating recovers. Older points systems cannot do this: once points are taken, they stay taken.

Repeat results do not stack up. Playing the same opponent fifty times does not earn fifty lots of points; it only makes the system more certain about which of the two is better. A frame won because the opponent left counts as a win, but not as a heavy one, since it was never played out.

How sure the system is. Each rating comes with a measure of uncertainty. Many frames against many different opponents give a confident rating. A few frames, or many frames against the same small group, leave it uncertain: the system knows how you compare with that group, but not with everyone else. Players are ranked on the cautious end of their likely range, so reaching the top takes a strong record against a wide range of opponents.

What comes next. At present a frame from last month counts as much as one from yesterday, so a player who has improved recently is recognised slowly, and a player who has stopped playing keeps their place. The planned change lets each player's level shift gradually over time, and makes the system less certain about anyone who has not played for a while. Their ranking then eases down until they play again.